

# Introduction to the Relational Model

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Based on Jennifer Widom slides

# Agenda

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The Relational Model

Relations

Tuples

Keys

# The Relational Model

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Proposed in 1969 by Edgar F. Codd

The most used model for databases

Very simple model

Query with high-level languages: simple yet expressive

Efficient implementations

# Relations

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Database = set of named **relations** (or **tables**)

Each relation has a set of named **attributes** (or **columns**)

The number of attributes is the arity of the relation

Student

| ID | name | GPA | photo |
|----|------|-----|-------|
|    |      |     |       |
|    |      |     |       |
|    |      |     |       |
|    |      |     |       |

College

| name | state | enroll |
|------|-------|--------|
|      |       |        |
|      |       |        |
|      |       |        |
|      |       |        |

# Tuples

Each **tuple** (or **row**) has a value for each attribute

No specific order between them

The number of tuples is the cardinality of the relation

Each attribute has a **type** (or **domain**)

Set of possible values. Examples: integer, text

College

| name     | state | enroll |
|----------|-------|--------|
| Stanford | CA    | 15,000 |
| Berkeley | CA    | 36,000 |
| MIT      | MA    | 10,000 |
|          | .     |        |
|          | .     |        |

Student

| ID  | name | GPA | photo |
|-----|------|-----|-------|
| 123 | Amy  | 3.9 | 😊     |
| 234 | Bob  | 3.4 | 😞     |
|     |      |     |       |
|     | .    |     |       |
|     | .    |     |       |

# Schema vs. Instance

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## Schema

structural description of relations in database  
name, attributes and types of those attributes  
typically set up in advance

## Instance

actual contents at given point in time  
change over time

# NULL value

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Special value for “unknown” or “undefined”

Useful but one must be careful when querying relations with NULL values

GPA>3.5; GPA≤3.5; GPA>3.5 OR GPA≤3.5

Student

| ID  | name  | GPA  | photo |
|-----|-------|------|-------|
| 123 | Amy   | 3.9  | 😊     |
| 234 | Bob   | 3.4  | NULL  |
| 345 | Craig | NULL | 😊     |
|     | .     |      |       |
|     | .     |      |       |

# Key (actually, candidate key)

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Set of one or more attributes whose combined values are unique within a relation

Often denoted by underlying the set of key attributes

## Importance

Identify specific tuples, efficiency, refer to tuples of another relation

Student

| <b>ID</b> | <b>name</b> | <b>GPA</b> | <b>photo</b> |
|-----------|-------------|------------|--------------|
| 123       | Amy         | 3.9        | ☺            |
| 234       | Bob         | 3.4        | ☹            |
|           |             |            |              |
|           | .           |            |              |
|           | .           |            |              |

Classroom

| <b>building</b> | <b>number</b> | <b>capacity</b> |
|-----------------|---------------|-----------------|
| B               | 001           | 184             |
| B               | 002           | 184             |
| I               | 001           | 50              |
|                 | .             |                 |
|                 | .             |                 |

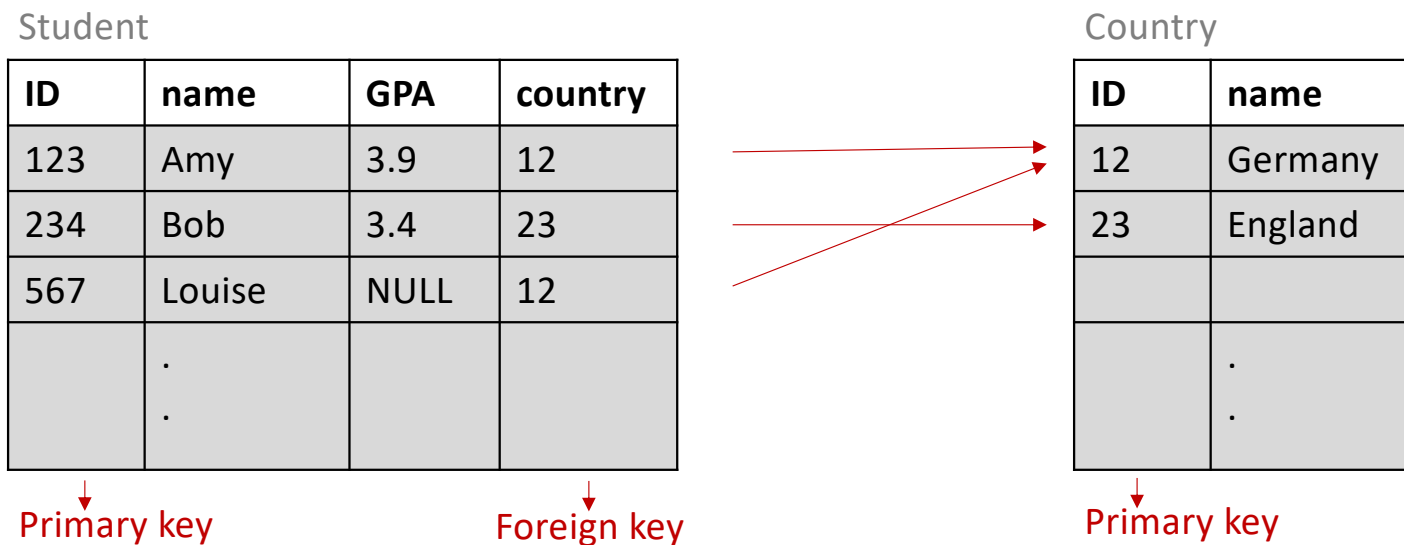


# Foreign Key

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An attribute (or set of attributes) that always matches a key attribute in the same or another relation

Often denoted by an arrow pointing to the name of the relation being referenced (see the Notation slide)



# Relational Notation

Student (ID, name, GPA, country->Country)

Classroom (building, number, capacity)

Country (ID, name)

In foreign keys, the name of the attribute can be different from the referenced attribute/relation

Country

| <u>ID</u> | name    |
|-----------|---------|
| 12        | Germany |
| 23        | England |
|           |         |
|           | .       |

Student

| <u>ID</u> | name | GPA | country |
|-----------|------|-----|---------|
| 123       | Amy  | 3.9 | 12      |
| 234       | Bob  | 3.4 | 23      |
|           |      |     |         |
|           | .    |     |         |

Classroom

| <u>building</u> | <u>number</u> | capacity |
|-----------------|---------------|----------|
| B               | 001           | 184      |
| B               | 002           | 184      |
| I               | 001           | 50       |
|                 | .             |          |

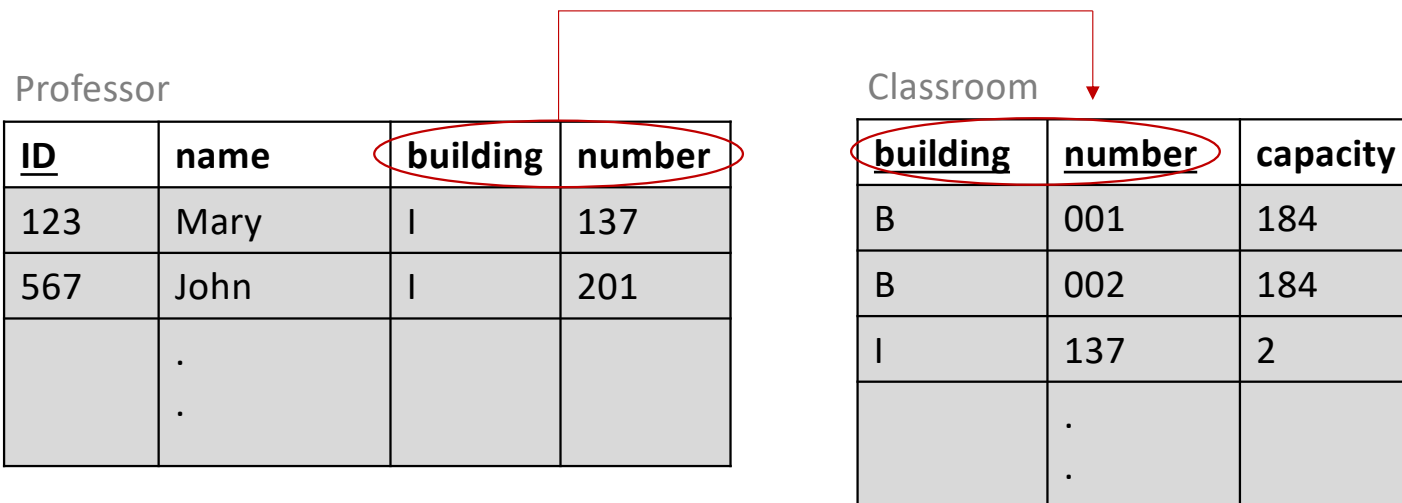
# Composite Keys

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A composite key is a multi-column primary-key or foreign-key

Classroom (building, number, capacity)

Professor (ID, name, building->Classroom.building, number->Classroom.number)



# Readings

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Jeffrey Ullman, Jennifer Widom, A first course in Database Systems 3<sup>rd</sup> Edition

Section 2.1 – Basics of the Relational Model